Performance of multivariable risk prediction algorithms in predicting future COPD exacerbations: a population-based study

Background

Guidelines stratify exacerbation risk in COPD by exacerbation history alone: a crude rule that gives the patient no number to act on. We externally validated the Acute COPD Exacerbation Prediction Tool (ACCEPT) in routine UK primary care, updated it, and asked: does a multivariable model beat the current standard of care?

Methods

- CPRD Aurum: UK primary-care records from 1,491 practices, linked to Hospital Episode Statistics, 2004–2020
- Adults ≥40 years with diagnosed COPD
- Index date: one randomly selected annual COPD review, at least 14 days clear of any exacerbation
- Outcome: ≥1 moderate/severe exacerbation within 12 months (secondary: ≥1 severe)
- Four strategies compared: any exacerbator history, frequent exacerbator history, ACCEPT 2.0, recalibrated ACCEPT 3.0-UK
- Evaluated on discrimination, calibration and net benefit

Study population

- 158,384 patients with COPD
- 55.0% male · mean age 71.5 years
- Mean FEV_1 60.4% predicted · mean mMRC 2.6
- 42.5% exacerbated within 12 months

Robustness

- Consistent across sex, smoking status, ethnicity, deprivation and region, and in spirometry-confirmed COPD (n=73,304) and complete cases (n=36,664)

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ERS abstract

A multivariable risk model predicts COPD exacerbations **significantly better than exacerbation history** alone in UK primary care

WHAT THE MODEL NEEDS

MANDATORY: the model will not run without these

Age	Sex	FEV_1 % predicted	mMRC dyspnoea score	Moderate exacerbations past 12 months	Severe exacerbations past 12 months

OPTIONAL: estimates are more precise when these are recorded: BMI · current smoking · home oxygen past year · statins · LAMA · LABA · ICS

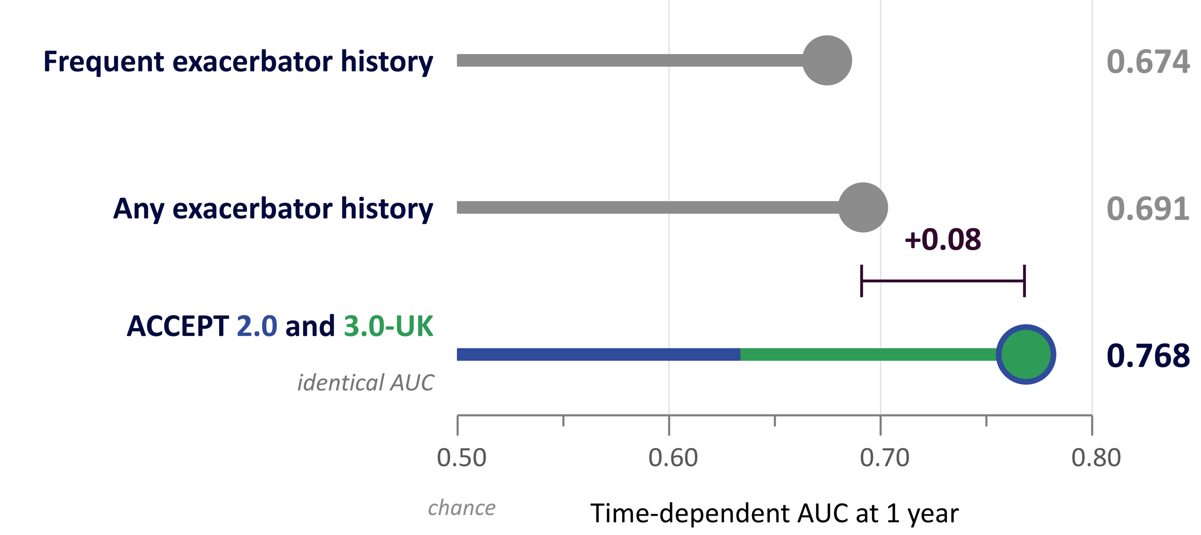
12-MONTH OUTPUT

Primary: risk of ≥1 moderate or severe exacerbation
Secondary: risk of ≥1 severe exacerbation

Moderate: antibiotics + oral corticosteroids for 5–14 days, ≥2 respiratory symptoms with either drug, an LRTI code, or an AECOPD code.
Severe: hospitalisation with COPD as the primary diagnosis.

1 Discrimination: AUC at 1 year

Both ACCEPT versions share one AUC: recalibration rescales risks without reordering patients.



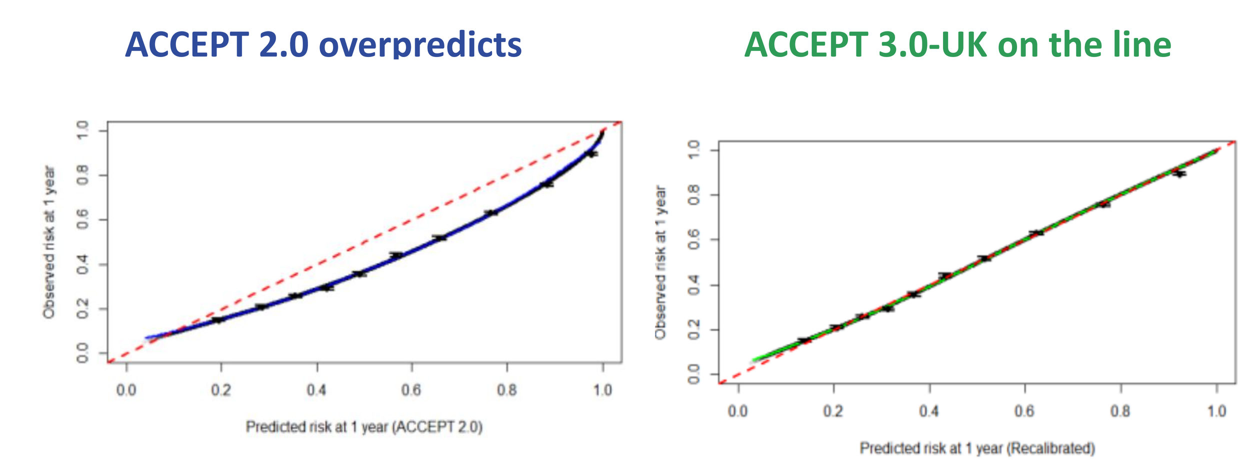
Severe-exacerbation outcome: AUC 0.774 (0.770–0.779)

Sharper risk stratification

Exacerbation history sorts patients into two crude bins ACCEPT ranks them markedly better with **AUC 0.77** and returns a risk each patient can act on.

2 Calibration:

ACCEPT 2.0 sat below the diagonal across the whole risk spectrum. ACCEPT 3.0-UK tracks it.



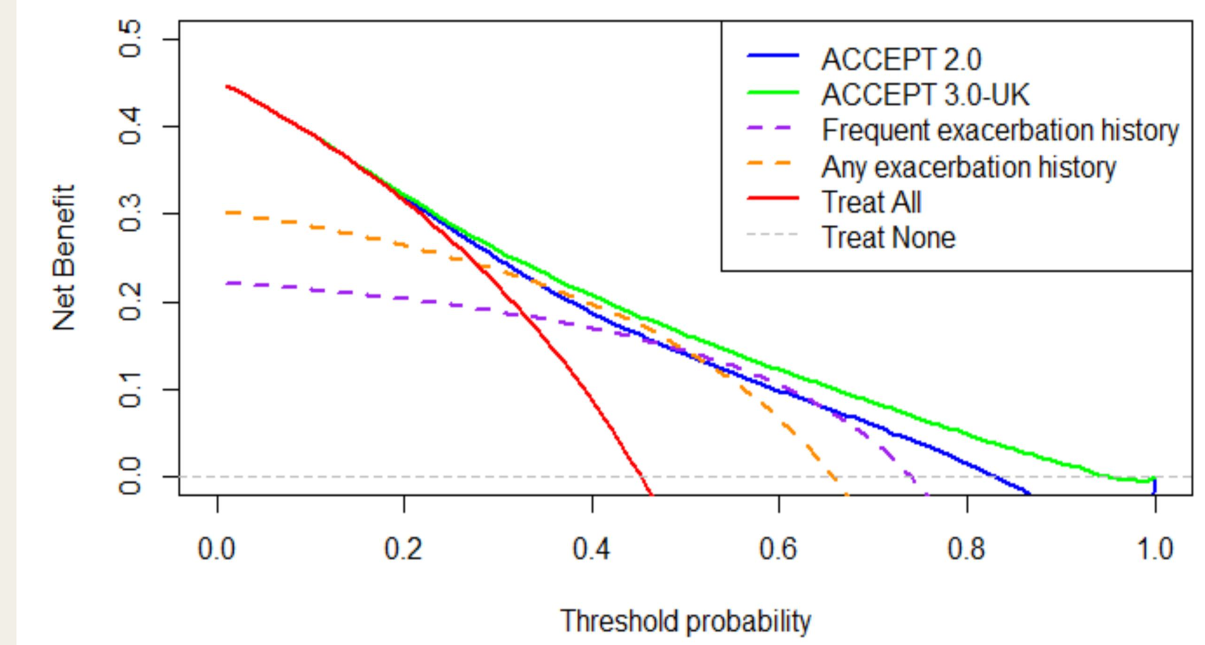
O/E 0.81 → O/E 1.00

Recalibration was essential

ACCEPT 2.0 overpredicted 12-month risk by **19%** (O/E 0.81), and by more than **twofold** for severe exacerbations (O/E 0.47). ACCEPT 3.0-UK restored O/E and slope to **1.00**, leaving discrimination untouched.

3 Clinical utility:

ACCEPT 3.0-UK (green) sits above every alternative from roughly an 11% threshold upward.



Highest net benefit of the four

Across the plausible treatment-threshold range, ACCEPT 3.0-UK was the **most clinically useful** of all four strategies. It is implemented in the open-source ***accept*** R package.

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Mehareen J, et al. Thorax 2026. doi:10.1136/thorax-2026-224814. Code: github.com/resplab/papercode/tree/main/ACCEPT3-UK. Model: github.com/resplab/accept. Funded in part by CIHR Team Grant PHT 178432; data access funded by the Respiratory Effectiveness Group. Based on CPRD data obtained under licence from the UK MHRA; interpretation and conclusions are the authors' alone.

Full paper
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